

Programme	BSc Mathematics Honours			
Course Code	MAT8VN401			
Course Title	INTRODUCTION TO ARTIFICIAL INTELLIGENCE			
Type of Course	Vocational Minor – Introduction to AI			
Semester	VIII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial	Practical	Total Hours
		per week	per week	
	4	3	2	75
Pre-requisites	Python Programming, Foundation of Mathematics, Machine Learning			
Course Summary	This course on "Introduction to Artificial Intelligence" offers a thorough exploration of AI fundamentals and techniques. Covering topics like representation, search algorithms, and intelligent agents, students' progress to advanced concepts including knowledge representation, neural networks, and practical implementations. With hands-on sessions focusing on algorithm implementation and machine learning models, students gain both theoretical understanding and practical skills essential for AI development.			

Course Outcome

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand foundation principles, mathematical tools and program paradigms of AI and Apply problem solving through search for AI applications	U	C	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
CO2	Understand formal methods of knowledge representation and Apply logic and reasoning techniques to AI applications	U	P	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
CO3	Apply intelligent agents for Artificial Intelligence programming techniques	Ap	P	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Module	Unit	Content	Hrs (45 +30)	Ext. Marks (70)
I		Introduction to Artificial Intelligence	10	Min.15
	1	Introduction to AI, History and Evolution of AI, Applications		
	2	Introduction to representation and search		
	3	The Propositional calculus, Predicate Calculus, Calculus expressions and Applications		
	4	State Space Search, Production Systems, Problem Characteristics, types of production systems, Graph theory		
	5	Intelligent Agents: Agents and Environments, The nature of environments, The structure of agents. concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation		
II		Search Strategies	14	Min.15
	6	Uninformed Search Strategies - Breadth First Search, Depth First Search, Depth Limited Search, Iterative Deepening Depth First Search		
	7	Bidirectional Search, Comparison of Uninformed search Strategies, Searching with partial information		
	8	Sensor-less problems, Contingency problems		
	9	Informed Search Strategies - Generate& test, Hill Climbing, Best First Search		
	10	A* and AO* Algorithm, Constraint satisfaction, Backtracking Search		
	11	Game playing: Minimax Search, Alpha-Beta Cutoffs		
	12	Optimal Decisions in Games, Stochastic Games		
III		Knowledge Representation	13	Min.15
	13	Knowledge Representation -Knowledge based agents, Wumpus world		
	14	Knowledge Representation -issues, The frame problem.		
	15	First order Logic: Representation, Inference, Reasoning Patterns, Resolution, Forward and Backward Chaining		

	16	Propositional Logic: Representation, Inference, Reasoning Patterns, Resolution, Forward and Backward Chaining		
	17	Agent based and distributed problem solving		
	18	Introduction to Expert System Technology, Bayes Rule, Bayesian Network, Hidden Markov Model, Decision Network		
IV		Introduction to ANN	8	Min.15
	19	Introduction ANN, biological neuron, Artificial neuron		
	20	Perceptron Learning		
	21	Back Propagation algorithm		
	22	Introduction to Natural Language Processing, Pattern recognition Case study - Enhancing Customer Service with AI-Powered Chatbots		
		Practical's	30	
	1	Write a program to implement depth first search algorithm.		
	2	Write a program to implement breadth first search algorithm.		
	3	Write a program to simulate 4-Queen / N-Queen problem.		
	4	Write a program to solve tower of Hanoi problem.		
	5	Write a program to implement alpha beta search.		
	6	Write a program for Hill climbing problem.		
	7	Write a program to implement A*algorithm		
	8	Write a program to implement AO*algorithm		
	9	Design the simulation of tic-tac-toe game using min-max algorithm		
	10	Write a program to shuffle Deck of cards		
	11	Write a program to derive the predicate.		
	12	Solve constraint satisfaction problem (a) Derive the expressions based on Associative law		

		(b)Derive the expressions based on Distributive law.		
	13	Develop a simple text-based game using Python that simulates a classic "Guess the Number" game. The game should generate a random number between 1 and 100 and prompt the player to guess the number. After each guess, the game should provide feedback to the player (e.g., "Too high", "Too low", or "Correct!") and keep track of the number of attempts it takes for the player to guess the correct number. Once the player guesses the correct number, the game should display the number of attempts and ask if the player wants to play again		
	14	Train a simple machine learning model, such as a linear regression or logistic regression classifier, using a dataset of your choice and evaluate its performance using appropriate metrics.		
	15	Implement a decision tree classifier from scratch and apply it to a classification task with a real-world dataset		
		References		
	1	S. Russel and p. Norvig, Artificial intelligence – A Modern Approach, 3rdEdn, Pearson		
	2	Artificial Intelligence: Elaine Rich, Kevin Knight, McGrawHill		
	3	Artificial Intelligence by Luger (Pearson Education)		
	4	D W Patterson, introduction to Artificial Intelligence and Expert Systems, PHI, 1990		
	5	Deep Learning by Ian Goodfellow, Yoshua Bengio, and Aaron Courville:		

Note: Proofs of all the results are exempted for the end semester exam.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	1	3	2	3	2	3	1	2
CO 2	3	2	3	2	3	2	3	1	2
CO 3	2	3	3	2	3	2	3	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓